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DT081A	T101	2025-01-07
Kursnamn	Datateknik AV, Signal- och bildbehandling	
Provnamn	Tentamen	
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Ämne		

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2025-01-07. Examination in

DT081A – Signal and Image Processing, 6 hp

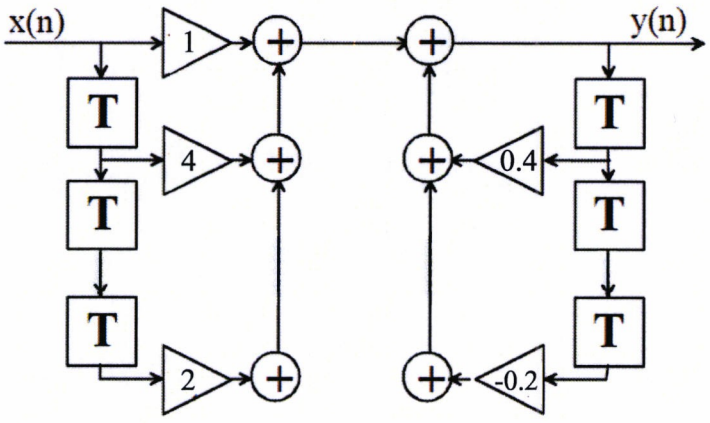
Allowed aids	Dictionary, non-programmable calculator and drawing tools (e.g ruler), Physics Handbook, TeFyMa, and formula collection booklet for mathematics/physics courses at MIUN. A collection of formulas is also presented at the end of this exam.
Miscellaneous	Don't forget to: 1. Submit one, or several pages, per solved assignment and NOT several solved assignments per page. 2. Order the handed in pages with the solved assignments in ascending order. 3. Describe and clearly motivate all solutions, assumptions, programming constructs, etc.. 4. Attach all neatly drawn figures that you might want to refer to in a solution. **Failing to follow the above instructions will render point reduction.**

Good Luck!

1. What does the following signal processing concepts mean? Be clear, concise, and relevant.

a) Convolution (1p)
b) Transfer function (1p)
c) Impulse response (1p)
d) Sampling (1p)
e) Aliasing (1p)

2. A time discrete system is defined as below.



1

a) Describe the system based on its impulse response behavior. (1p)
b) Present the system's difference equation. (2p)

3. Given a discrete-time system described by the difference equation

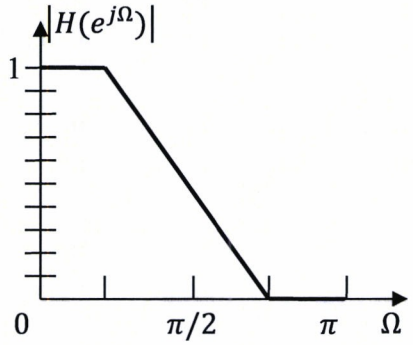
$$y(n) = x(n) + y(n-1) + y(n-2)$$

with the input signal

$$x(n) = \delta(n)$$

a) Identify the type of response that the system produces by the given input $x(n)$. (1p)
b) Calculate the first 5 samples of the output $y(0), y(1), \ldots, y(4)$. (2p)
c) The discrete time fourier transform (DTFT) of $y(n)$ can be found to be
$Y(e^{j\Omega}) = X(e^{j\Omega}) + e^{-j\Omega}Y(e^{j\Omega}) + e^{-2j\Omega}Y(e^{j\Omega})$,
which gives the system's frequency function
$H(e^{j\Omega}) = \frac{Y(e^{j\Omega})}{X(e^{j\Omega})} = \frac{1}{1-e^{-j\Omega}-e^{-2j\Omega}} = \frac{e^{2j\Omega}}{e^{2j\Omega}-e^{j\Omega}-1}$.
Evaluate the amplitude magnification $|H(e^{j\Omega})|$ for frequency $\Omega = \frac{\pi}{2}$. (3p)

4. A signal $s(t)$ have been sampled using $f_s = 100$ kHz. The amplitude spectrum of the sampled signal is shown in the figure below for a limited normalized frequency range.

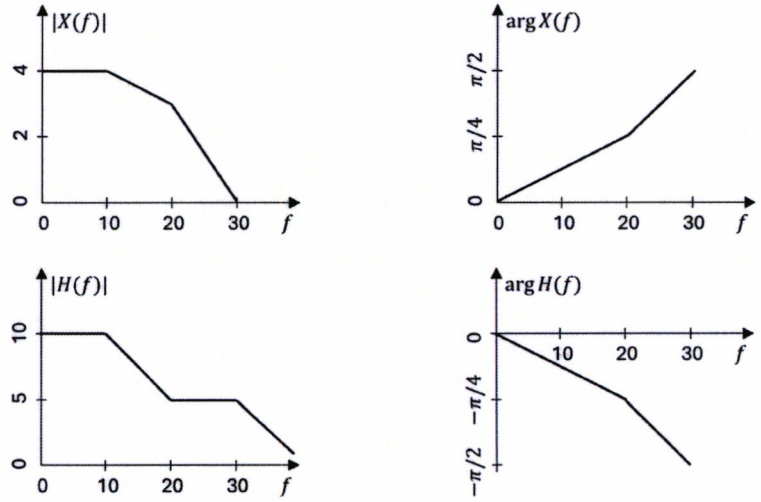


a) Sketch the amplitude spectrum for an extended normalized frequency range by extending axis to $-2\pi < \Omega < 2\pi$. (2p)
b) What is the lowest sampling frequency in Hz that the signal $s(t)$ could be sampled in while still being possible to be reconstructed exactly? Motivate. (3p)

5. An LTI system $H(f)$ is defined by its amplification and phase-shift below. A signal $x(t)$ is fed to the system, giving the output $y(t)$.

a) Give the mathematical relation between amplitudes and phases of the input and the output. (2p)

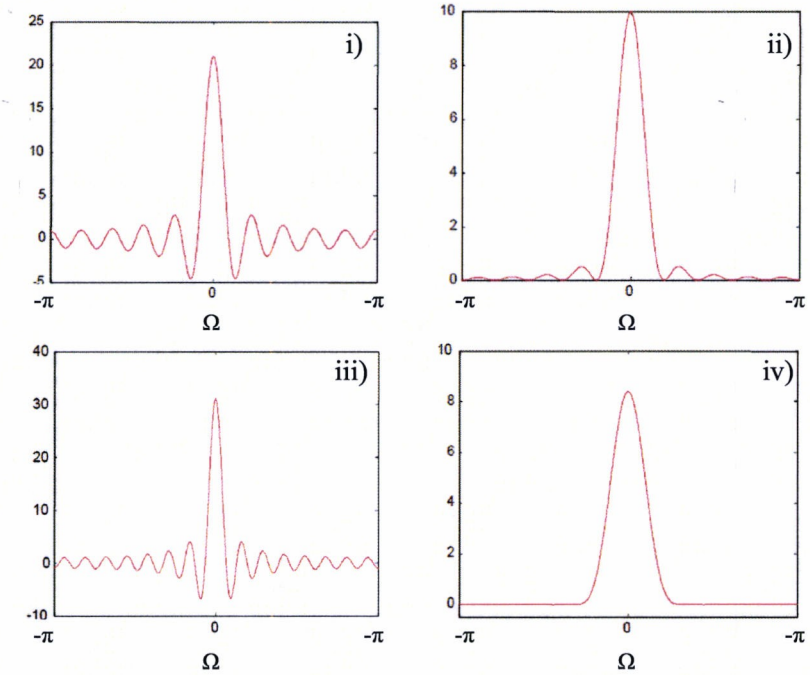
b) Draw the amplitude spectrum and phase spectrum for the output $y(t)$. (4p)



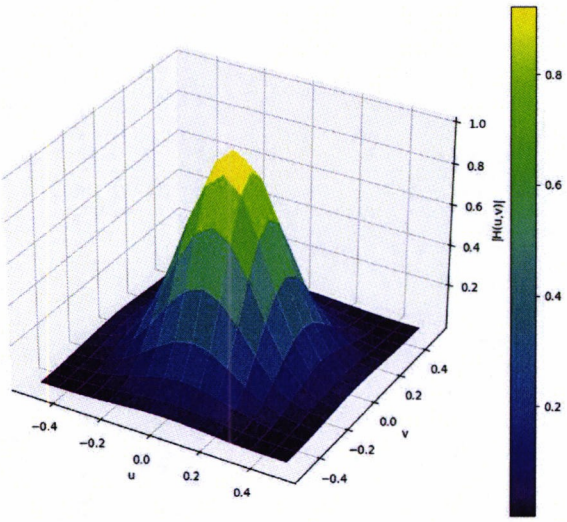
6. The plots below show the frequency spectra (i to iv) of different window functions with varying window lengths employed in spectrogram analysis.

a) Based on the frequency spectra, which window function achieves the best frequency separation? Justify your choice. (2p)

b) Which window function best minimizes spectral leakage (where unwanted frequency components appear away from the main frequency)? Motivate. (2p)

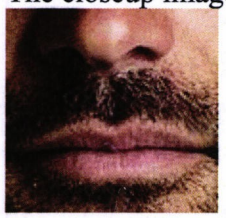


7. A two-dimensional linear system can partially be described by the figure below. Based on the information in the figure, describe the system's frequency selective properties.



(3p)

8. The closeup image below has been filtered with 4 different filters.



Match the following kernels (1–5) with the corresponding filtered results (i–v). Justify your choices by explaining how the filter kernels affect the image.

Kernels

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$ **1)**

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$$ **2)**

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$ **3)**

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$ **4)**

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$ **5)**

Filter results



i) **ii)** **iii)** **iv)** **v)**

(5p)

9. Given the two matrices representing the input $x(i,j)$ and output $y(i,j)$ of a linear 2D filter: describe the $3x3$ point spread function (PSF) or kernel $h(i,j)$ that transforms $x(i,j)$ into $y(i,j)$. Assume boundary conditions are zero, meaning pixels outside the matrix boundaries of $x(i,j)$ and $y(i,j)$ are zero.

$x(i,j)$

i \ j	0	1	2	3	4	5
0	0	0	0	0	0	144
1	0	112	0	0	0	0
2	0	0	0	224	0	0
3	80	0	64	0	0	0
4	0	0	0	176	0	0
5	0	0	0	0	0	48

$y(i,j)$

i \ j	0	1	2	3	4	5
0	0	14	0	0	18	72
1	14	56	14	28	0	18
2	10	14	36	112	28	0
3	40	18	32	58	0	0
4	10	0	30	88	22	6
5	0	0	0	22	6	24

(6p)

Collection of Formulas

Base Signals

$$x(t), (continous)$$

$$x[n], (discrete)$$

Stochastic

$$w(n) \sim \mathcal{N}(0, \sigma^2)$$

Linearity

$$ax_1 + bx_2 \rightarrow ay_1 + by_2$$

Time-Shift

$$x(t - t_0) \rightarrow y(t - t_0)$$

Causality

$$h(t) = 0,\ t < 0$$

Nyquist-Shannon Sampling Theorem

$$f_s \geq 2f_{max}$$

Convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau)d\tau \text{ (continuous)}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k] \text{ (discrete)}$$

Euler's Formula

$$e^{jx} = \cos(x) + j\sin(x)$$

Fourier Series

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

DTFT

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

DFT

$$X(k) = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

Time/Space Shift Property

$$x(t - t_0) \leftrightarrow X(\omega) e^{-j\omega t_0} \text{ (continuous)}$$

$$x[n - n_0] \leftrightarrow X(e^{j\Omega}) e^{-j\Omega n_0} \text{ (discrete)}$$

Spectrogram

$$S(t, \omega) = |X(t, \omega)|^2$$

FIR

$$y[n] = \sum_{k=0}^{M} b_k x[n - k]$$

IIR

$$y[n] = \sum_{k=1}^{N} a_k y[n - k] + \sum_{k=0}^{M} b_k x[n - k]$$

Frequency Response

$$H(e^{j\Omega}) = |H(e^{j\Omega})| e^{j \arg H(e^{j\Omega})}$$

PSF Basic

$$h(x, \alpha, y, \beta)$$

Shift-Invariant

$$g(\alpha, \beta) = \sum_{x} \sum_{y} f(x, y) h(\alpha - x, \beta - y)$$

DCT Basis

$$U_{x\alpha} = \sqrt{\frac{2}{N}} \cos\left(\frac{\pi(2x + 1)\alpha}{2N}\right)$$

RGB to YCbCr

$$\begin{bmatrix} Y' \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Gamma Correction

$$C' = C^{1/\gamma}$$